POSTS AND TELECOMMUNICATIONS
INSTITUE OF TECHNOLOGY
INFORMATION TECHNOLOGY

# INTERNSHIP

## REPORT

# WoxionChat

Name: NGUYEN PHAM TRUNG HIEU
Student ID: B23DCCE031
Class ID: B23CQCE04-B
Instructor: KIM NGOC BACH

Hanoi

# Table of contents

**1 Overview** **2**

**2 Technical Accomplishments** **2**
2.1 Intensive SFT & Dataset Augmentation . . . . . 2
2.2 Performance Infrastructure & Efficiency . . . . . 2
2.3 Agentic Evolution . . . . . 2

**3 Technology Stack Update** **3**

**4 Progress Evaluation** **3**

**5 Upcoming Milestone (Next Week)** **3**

# Table of contents

# 1 Overview

Building upon the previous VMLU benchmarks, this week's development cycle was dedicated to extending the **Supervised Fine-Tuning (SFT)** phase. We prioritized high-quality data curation and iterative training to address specific edge cases in Vietnamese business communication. The objective remains to solidify the model's core instruction-following capabilities before transitioning to preference-based alignment.

# 2 Technical Accomplishments

## 2.1 Intensive SFT & Dataset Augmentation

The SFT process was refined to focus on accuracy and stylistic consistency:

- **Data Quality Audit:** Conducted a comprehensive review of the existing SFT dataset, removing low-quality samples and correcting linguistic inconsistencies in the "Staff" vs. "Boss" interaction scenarios.
- **Edge-Case Training:** Integrated 5,000+ new high-quality instruction-response pairs focusing on complex administrative tasks, error handling, and multi-turn workplace coordination.
- **Linguistic Normalization:** Fine-tuned the model to better handle regional Vietnamese dialects and professional terminology commonly used in corporate environments.

## 2.2 Performance Infrastructure & Efficiency

- **Context Compression:** Researched and implemented a lightweight context compression technique, allowing the model to "remember" longer conversation threads without exponentially increasing GPU memory usage.
- **Inference Bottleneck Analysis:** Conducted a deep-dive into the AI API latency. By refactoring the model-loading sequence in the Django backend, we achieved a further 10% improvement in concurrent request handling.

## 2.3 Agentic Evolution

- **Multi-Step Reasoning Refinement:** Improved the agent's internal "Chain of Thought" (CoT) to handle nested tool-calling (e.g., "Find the employee's last session, then check their IP, and finally generate a summary report").
- **Consistency Checks:** Added an automated validation layer that cross-references AI-generated tool parameters against the MongoDB schema before execution.

# 3 Technology Stack Update

- **SFT Framework:** PEFT (Parameter-Efficient Fine-Tuning) using QLoRA for efficient resource utilization.
- **Monitoring:** Integrated Prometheus and Grafana for real-time tracking of AI API response times and GPU utilization during training iterations.
- **Backend:** Optimized Django-Celery task serialization for faster data transfer between the API and the model worker.

# 4 Progress Evaluation

- **Status:** On Track.
- **Observations:** Continuing with SFT has proven beneficial as the model's stability in following complex, multi-step instructions has increased by approximately 25%. The responses are now more predictable and align closely with the defined corporate personas.

# 5 Upcoming Milestone (Next Week)

1. **SFT Finalization:** Concluding the final round of SFT based on the latest data quality audits.
2. **Closed Beta Launch:** Deploying the system to a limited group of internal "Boss" users for real-world feedback on the fine-tuned model.
3. **Initial Alignment Planning:** Preparing the infrastructure for DPO (Direct Preference Optimization) based on the feedback gathered from the beta testers.